

Problem

Calculate the rms value for each of the following functions:

(a) $i(t) = 10 \text{ A}$

(b) $v(t) = 4 + 3 \cos 5t \text{ V}$

(c) $i(t) = 8 - 6 \sin 2t \text{ A}$

(d) $v(t) = 5 \sin t + 4 \cos t \text{ V}$

Step-by-step solution

Step 1 of 4

(a)

$$I_{rms} = 10 \text{ A}$$

Step 2 of 4

(b)

$$V_{rms}^2 = 4^2 + \left(\frac{3}{\sqrt{2}}\right)^2$$

$$V_{rms} = \sqrt{16 + \frac{9}{2}}$$

$$\therefore V_{rms} = 4.528 \text{ V}$$

Step 3 of 4

(c)

$$I_{rms} = \sqrt{64 + \frac{36}{2}}$$

$$\therefore I_{rms} = 9.055 \text{ A}$$

Step 4 of 4

(c)

$$V_{rms} = \sqrt{\frac{25}{2} + \frac{16}{2}}$$

$$\therefore V_{rms} = 4.528 \text{ V}$$